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Social robots for English language teaching

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Introduction

As a result of recent fast-growing technological developments, providing educational resources through which learners can interact with human-like interlocutors has become more available and affordable (van den Berghe et al. 2019; Randall 2019). Among these resources, social robots have gained increasing attention in the field: not only can these physically interact with learners in a socially acceptable fashion (Bartneck and Forlizzi 2004; Han 2012) but they also have become more accessible due to ever-lower costs (e.g. renting NAO [Figure 1] for a month costs around £350 as of April 2021). They have been suggested to 'allow for more natural interaction' (van den Berghe et al. 2019: 260) and have been found to be more useful, reliable, and interesting than other forms of technology (Kidd and Breazeal 2004; Wainer et al. 2007). Nevertheless, social robots appear to be a relatively new technology for most ELT professionals and remain underused in language lessons and understudied in the ELT research community. Thus, this article aims at introducing social robots for ELT purposes, summarizing the findings of previous ELT studies on this technology, and presenting different ways of employing social robots in ELT lessons as well as caveats for ELT practitioners and directions for future research.

What is a social robot?

As its name indicates, a social robot is a type of robot designed to 'socially' interact with people. This quality makes it different from, for example, an industrial robot used for manufacturing. Being a 'robot', it also has a physical body, which can share physical space with learners, and thus it is different from virtual characters designed to interact with learners only on screens, as, for example, chatbots do (cf. Lee et al. 2020). In view of such a strength, social robots have been widely applied in various educational fields (e.g. Leyzberg et al. 2012; Kennedy et al. 2015), with ELT not being an exception (e.g. Tsai 2019; Vogt et al. 2019).

Figure 1 illustrates one example of a social robot, NAO, which was developed by SoftBank Robotics. NAO is equipped with seven important features for his basic movements (battery, sonar, cameras) as well as his verbal (microphones, speakers) and non-verbal (eyes, hands) interactions with learners.

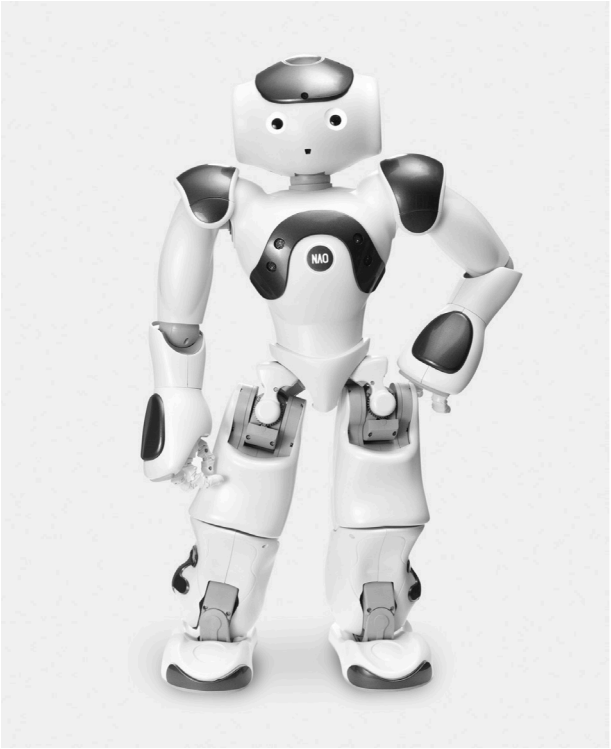


Figure 1 SoftBank Robotics' small bipedal robot NAO (https://www.softbankrobotics.com/corp/robots/). ©SoftBank Robotics

By virtue of their physical embodiment, social robots have been suggested to provide language learners with direct and physical interactions (Han 2012), which is considered one of the basic ingredients for language learning (Iverson 2010). In addition, as social robots are generally humanoids or anthropomorphized animal shapes, they have been valued for their ability to serve as familiar conversational partners (Duffy 2003), having potential to lower the affective filter of language learners (Han 2012).

Application of social robots in ELT: lessons from research

In this section, we summarize the findings of previous ELT studies on how social robots have been applied to ELT classrooms. To date, we have identified a dozen studies that assessed the effects of employing social robots on English learning, whereas another dozen examined their effects on the changes in affective status among English learners.

In brief, the overall findings from the previous studies were as follows. First, social robots seem to contribute positively to changing affective variables such as motivation and engagement (e.g. Hong et al. 2016; Alemi, Meghdari, and Haeri 2017). Social robots have been suggested to be particularly effective for lowering learners' anxiety when they engage in English learning activities. Second, in regard to English learning outcomes, the previous studies have reported that social robots could bring about positive outcomes related to English learning achievements (e.g. Tanaka and Matsuzoe 2012; van den Berghe et al. 2018). More specifically, involving social robots led to statistically similar or even higher outcomes compared with traditional ELT contexts (i.e. no social robot).

This positive effectiveness of social robots in English teaching may be intriguing enough to make ELT practitioners eager to explore how these resources have been actually applied in ELT classrooms, and we will introduce some possible ways to employ social robots for ELT purposes in terms of the different roles they could take.

It should be noted here that the following example applications of social robots in ELT classrooms are situated in English vocabulary learning contexts. Our literature search indicated that the majority of robot-assisted English learning research predominantly targeted vocabulary learning; this tendency may be related to the characteristics of vocabulary lessons, in which the target (lexical) items could be presented individually and independently, which social robots may be able to manage better than other, more complex skills. Nevertheless, this does not limit the potential of social robots because they can be used for teaching different skills as well. For example, in Alemi and Haeri (2020), teachers targeted pragmatic features such as 'thanking' speech acts, by creating a classroom situation in which learners brought a certain object to the robot, and the robot expressed gratitude to them. Robots could also be employed in a reading-aloud activity, in which they could read the target passage aloud to the learners, and also perform some movements which illustrate the scene described in the passage, as presented in Hong et al. (2016).

The first role that a social robot could take is that of a tutor, giving language lessons without the presence of a human teacher. For example, in the study by de Wit et al. (2018), primary school students were given one-on-one tutoring by a social robot, with the goal of the lesson being the acquisition of English words related to animal names. In this study, the students were first exposed to the target vocabulary via a laptop. Then the robot served as an English tutor by carrying out the vocabulary learning game. In doing so, the robot explained the procedure of the activity, gave the stimuli related to the target vocabulary, and offered positive and negative feedback—depending on the students' responses.

Social robots can also serve as learners' peers, with some of them being pre-programmed to be on a par with learners. In the study by van den Berghe et al. (2018), Dutch kindergarden learners were paired with a robot, which was programmed to engage in English word learning with a child using a tablet. Although the robot was considered the peer from the child's perspective, it also made comments on learners' manipulations or offered some help when the child had difficulty in the target task in order to boost the child's motivation and give some type of scaffolding. Learners could also practice daily expressions with robots, which serve as conversation partners (Alemi and Haeri 2020). In doing so, learners may be given a dialogue including previously learned expressions, and practice those expressions with robots, which are programmed to engage in such a conversational practice. In addition, these robots could be programmed to make mistakes when conversing with learners, which would help them realize that making mistakes is non-threatening, thereby reducing the learners' level of anxiety (Alemi et al. 2017).

Learners could also be situated in 'care-receiving' paradigm, in which they engage in teaching target language items to a robot. Such an idea was incarnated in Tanaka and Matsuzoe (2012), who examined Japanese pre-schoolers' acquisition of English verbs by having them teach these vocabulary items to the robot by acting out. It has been suggested that this paradigm would be beneficial for learners, because they may feel more 'comfortable expressing lack of understanding' and achieve a higher level of 'task motivation' in this 'learning-by-teaching' mode (Randall 2019: 9).

Another role of social robots that has received attention is that of a teaching assistant, serving as a teacher's 'sidekick', so to speak. The study by Alemi, Meghdari, and Ghazisaedy (2014) illustrates a lesson plan in which a target robot could assist in a teacher's vocabulary teaching. In their study, a robot called Nima not only presented the target English vocabulary and its uses in dialogues, but also engaged in practising vocabulary with the students, with the latter including Nima's checking of the students' comprehension. The teacher, while assigning the job of presenting the target vocabulary to Nima, oversaw Nima's interaction with the students and gave linguistic feedback and reinforcement (approving learners' answers and complimenting) to the students; employed additional multimedia (e.g. PowerPoint slides) to enhance students' English vocabulary learning; and intervened whenever necessary.

Although in its infancy, one technology which may be highly useful when employing social robots as a teaching assistant lies in personalization of feedback directed toward each learner (Randall 2019). In reality, it is almost impossible for a teacher to memorize each learner's progress and previous utterances. However, a social robot as a teaching assistant may engage

in conversing with each learners, store the aforementioned pieces of information, and give more personalized (linguistic) feedback to each learner, while the teacher is still in control of the overall lesson.

Caveats for teachers and directions for future research

Before applying social robots to ELT classrooms, we would like to suggest some caveats for ELT practitioners along with directions for future research. The first caveat is related to the limitation of automatic speech recognition (ASR) technology (Randall 2019). The ASR engine installed in social robots, in general, has not been developed to the extent that it could recognize EFL or ESL learners' (in particular younger ones) utterances to a satisfactory level. In this regard, future research is needed to collect and build up a large volume of corpora on EFL or ESL learners' utterances, which could be used to train the ASR engine for better comprehension of learners' utterances. Thus, a teacher may share such a characteristic of social robots with learners before introducing them, which may reduce learners' anxiety when robots do not appear to recognize their utterances.

The second caveat, which is frequently mentioned in the studies introduced in the previous section, is related to the novelty effect of introducing social robots to learners (van den Berghe et al. 2019), meaning that learners' interest in social robots may decline over time. ELT practitioners are encouraged to employ social robots in different ways by assigning different roles (see the section 'Application of social robots in ELT: lessons from research'), which may help learners to maintain their interests in this technology. In this regard, we need more longitudinal studies that employ social robots over an extended period of time, which may provide some answers regarding the degree of the aforementioned novelty effect.

Conclusions

This article aimed at introducing social robots for language teachers in view of previous ELT literature on this technology. In doing so, we presented some general characteristics of social robots as well as a summary of findings from previous ELT studies on this technology. We also offered some possible ways to apply social robots for ELT, which were followed by caveats for ELT practitioners and directions for future research. Although we are still far from having ideal social robots for ELT, we hope to see more research on this resource and its technological developments, which would initiate a new era of language learning, thereby making social robots an important addition to the ELT arsenal.

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